



**ARKA JAIN
University**
Jharkhand



Subject: Operating Systems

Course: B.Tech Branch: CSE SEM: IV

Assignment: 1

INTRODUCTION TO OPERATING SYSTEM

1. Compare the layered, monolithic, and microkernel structures of operating systems with examples. (*Understanding*)
2. Define an Operating System and list its primary functions. (*Remembering*)
3. Describe the services provided by an operating system and analyze their importance in system functionality. (*Analyzing*)
4. List and explain any two types of operating systems. (*Remembering*)
5. Explain the concept of a virtual machine and its significance in modern computing. (*Understanding*)
6. What are the different generations of operating systems? Provide a brief description of each. (*Remembering*)
7. Conduct a comparative study of UNIX and WINDOWS operating systems, highlighting their key differences and use cases. (*Analyzing*)

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Assignment: 2

PROCESSES AND THREADS

1. Define a Process Control Block (PCB). Explain the different types of schedulers in process scheduling. *(Remembering)*
2. What are the different states of a process, and how do process state transitions occur? Explain using a diagram. *(Understanding)*
3. Differentiate between pre-emptive and non-pre-emptive scheduling algorithms with examples. *(Understanding)*
4. Explain the benefits of threads and the concept of multithreading. *(Understanding)*
5. Describe the scheduling criteria: CPU utilization, throughput, turnaround time, waiting time, and response time. *(Analyzing)*
6. (I) Compare FCFS and SJF scheduling algorithms *(Analyzing)*
(II) Calculate the average turnaround time and average waiting time using FCFS scheduling for the given set of processes.

P_id	AT	BT
P1	3	1
P2	1	4
P3	4	2
P4	0	6
P5	2	3

7. What is context switching, and why is it necessary? *(Understanding)*
8. Define real-time scheduling and compare RM (Rate Monotonic) and EDF (Earliest Deadline First) scheduling techniques. *(Analyzing)*
9. For the given set of processes, calculate the average Turn Around Time and Average Waiting Time using Priority scheduling. *(Analyzing)*

P_id	AT	BT	Priority
P1	1	4	1
P2	2	5	2
P3	3	2	3
P4	4	1	4
P5	5	6	5
P6	6	3	6

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Assignment: 3

INTER-PROCESS COMMUNICATION & DEADLOCKS

1. Define Mutual Exclusion and explain its importance in inter-process communication. *(Remembering)*
2. What are race conditions, and how do they affect process execution? *(Understanding)*
3. Explain Peterson's Solution for achieving mutual exclusion with a diagram. *(Analyzing)*
4. Compare Semaphores and Monitors in terms of their use and working principles. *(Analyzing)*
5. Describe the Producer-Consumer Problem and explain how it can be solved using semaphores. *(Understanding)*
6. What are the necessary and sufficient conditions for a deadlock to occur? *(Remembering)*
7. (I) Explain the Banker's Algorithm

(II) Consider the following snapshot of a system:

Processes	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P ₀	1	1	2	4	3	3	2	1	0
P ₁	2	1	2	3	2	2			
P ₂	4	0	1	9	0	2			
P ₃	0	2	0	7	5	3			
P ₄	1	1	2	1	1	2			

- Calculate the content of need matrix.
 - Is the system in a Safe State? If yes, mention the sequence. *(Analyzing)*
8. Discuss the classical Dining Philosopher Problem and how it models deadlocks in systems. *(Understanding)*
 9. What is the difference between deadlock prevention and deadlock detection? Provide examples. *(Analyzing)*

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Assignment: 4

MEMORY MANAGEMENT

1. Define logical and physical addresses in memory management and explain their mapping. *(Remembering)*
2. What is internal and external fragmentation? Provide examples to differentiate them. *(Understanding)*
3. Explain the principle of paging with the help of a diagram showing logical to physical address translation. *(Understanding)*
4. What is demand paging? Describe the concept of virtual memory and its advantages in modern operating systems. *(Understanding)*
5. Explain the working of demand paging, including the handling of a page fault. *(Analyzing)*
6. (I) Compare the FIFO and LRU page replacement *(Analyzing)*
(II) Using FIFO page replacement scheme, calculate the page fault, miss ratio, and hit ratio for the given reference string: 4,7,6,1,7,6,1,2,7,2. Page frames=3
7. What is the working set in virtual memory, and how does it help in memory management? *(Understanding)*
8. Explain the structure and purpose of a page table with an example. *(Analyzing)*
9. What is a dirty page/dirty bit, and why is it significant in memory management? *(Remembering)*
10. What is Convoy effect? Using Optimal page replacement scheme, calculate the page fault, miss ratio, and hit ratio for the given reference string: 4,7,6,1,7,6,1,2,7,2. Page frames=4 *(Analyzing) (Create)*

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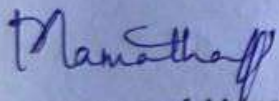
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Assignment: 5

I/O HARDWARE, FILE/DISK MANAGEMENT

1. What are the primary goals of I/O software, and how do device drivers play a role in achieving them? (*Understanding*)
2. Explain the working of Direct Memory Access (DMA) with a diagram and its importance in I/O operations. (*Understanding*)
3. Describe the structure of a disk and explain how it impacts disk scheduling. (*Understanding*)
4. Compare FCFS and SSTF disk scheduling algorithms using an example reference queue. (*Analyzing*)
5. What are the different file allocation methods, and how do they affect file system performance? (*Analyzing*)
6. Define free-space management in file systems and explain the bit vector and linked list approaches. (*Remembering*)
7. Discuss the linear list and hash table methods for directory implementation, highlighting their advantages and disadvantages. (*Understanding*)
8. A disk with 200 tracks (0-199) has the following disk queue for input/output requests: 120, 180, 30, 90, 150, 10, 70, 190. The initial head position is 50, and it moves toward the right-hand side. Calculate the total number of track movements for both the SCAN and C-SCAN algorithms. (*Analyzing*)
9. What is a boot block, and how does it contribute to system startup? (*Remembering*)
10. What is the role of device controllers in I/O hardware? (*Remembering*)


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